



SISMO SFTP FILE UPLOADER

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SftpClient Setup and Usage Guide - Ubuntu

This guide explains how to prepare, configure, and run SftpClient.jar on Ubuntu. It also includes instructions for scheduling the application to run daily at 05:00 AM and for removing the scheduled task.

Note: 05:00 AM is configured in the local time zone. It is recommended to adjust the schedule so that file uploads occur at 5:00 AM UTC.

Note: SftpClient is a Java-based application that connects to an SFTP server and uploads files to a specified directory.

Single-User and Multi-User Modes

SftpClient supports two operational modes:

1. Single-User Mode

- **Default mode** when only one username is provided in the SFTP_USER variable
- All files are uploaded using a single SFTP account
- Files are taken directly from the main directories (user_indicator, macro_indicator, portfolio)

2. Multi-User Mode

- **Activated automatically** when multiple comma-separated usernames are provided in SFTP_USER
- Files are distributed to and uploaded from user-specific subdirectories
- Each user's files are uploaded using their respective SFTP credentials
- **Important:** User-specific subdirectories are created automatically with the same names as the SFTP usernames

How Multi-User Mode Works:

1. When multiple users are detected, the application creates user-specific subdirectories in each main directory
2. Files from the main directories are copied to each user's subdirectory
3. Files are uploaded separately for each user using their respective usernames and the same private and public keys
4. **Directory structure is automatically created** - if directories don't exist, they will be created during execution



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Setting Up Multi-User Mode:

Update the SFTP_USER environment variable in run_macos.command with comma-separated usernames:

SFTP username(s) for authentication

For single user: export SFTP_USER=username

For multiple users: separate with commas, e.g., "user1,user2,user3"

Example: export SFTP_USER="user1,user2,user3"

export SFTP_USER=

⚠ Important: The subdirectory names **MUST** match exactly with the usernames specified in the SFTP_USER variable.

Prerequisites

1. **Java Runtime Environment (JRE)** Ensure the jre folder is included in the same directory as the SftpClient.jar.
2. **Environment Variables** The run_ubuntu.sh script requires several environment variables to be set. Update these to match your environment before execution.
3. **Required Folders** The following folders in the root directory of the application (created during the zip packaging) are used by default for file uploads:
 - portfolio: for portfolio files.
 - user_indicator: for stock-related indicator files.
 - macro_indicator: for macro indicator files.

You may create your own folders and update the environment variables in run_ubuntu.sh accordingly.

Environment Variables

Below there are key environment variables, their purposes, and usage examples:

SFTP_SERVER

- **Purpose:** Specifies the hostname or IP address of the SFTP server to which files will be uploaded.
- **Example:** sftp.example.com or 192.168.1.100

SFTP_USER

- **Purpose:** The username(s) used for authenticating with the SFTP server.
- **Single-User Example:** sftpuser
- **Multi-User Example:** user1,user2,user3

When multiple users are specified:

- User-specific subfolders can be automatically created in user_indicator, macro_indicator, and portfolio directories or can be created manually.



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- Each subfolder name must match the corresponding username exactly
- Files from main directories are distributed to all user directories
- The application uploads files for each user with their respective username and the same private/public keys

SFTP_PRIVATE_KEY

- **Purpose:** Path to the private SSH key file used for SFTP authentication.
- **Example:** /home/youruser/.ssh/id_rsa

SFTP_PHRASE

- **Purpose:** Passphrase for the private SSH key, if required. Leave empty if the key is not encrypted.
- **Example:** yourSecretPassphrase
(Leave blank if not needed)

SFTP_LOCAL_USER_INDICATOR_DIRECTORY

- **Purpose:** Directory path containing user indicator files to be uploaded.
- **Example:** /home/youruser/sftpclient/user_indicator

SFTP_LOCAL_MACRO_INDICATOR_DIRECTORY

- **Purpose:** Directory path containing macro indicator files to be uploaded.
- **Example:** /home/youruser/sftpclient/macro_indicator

SFTP_LOCAL_PORTFOLIO_DIRECTORY

- **Purpose:** Directory path containing portfolio files to be uploaded.
- **Example:** /home/youruser/sftpclient/portfolio

SFTP_PORTFOLIO_FILE_MAPPER

- **Purpose:** Maps external IDs to substrings in file names for portfolios, which are stored on a user's side, enabling, enabling automatic renaming before upload. There should be no hyphens/dashes in the portfolio names (e.g. Portfolio1 as below)
- **Example:** *PortfolioExternalId1-Portfolio1, PortfolioExternalId2-Portfolio2*
- **Mock File Name:** Portfolio1_Report.csv renamed to PortfolioExternalId1-[timestamp]-[operation].csv (The substring Portfolio1 in the file name matches the mapping PortfolioExternalId1-Portfolio1. The file is renamed to PortfolioExternalId1-[timestamp]-[operation].csv before being uploaded)

SFTP_PORTFOLIO_FILE_MAPPER_DEFAULT_OPERATION

- **Purpose:** Specifies the default operation for portfolio files.
 - F = replace file and delete history
 - M = modify file
- **Example:** M Files will be uploaded with the “modify” (update) operation unless specified otherwise.



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SFTP_USER_INDICATOR_FILE_MAPPER

- **Purpose:** Same logic as for portfolios, but applies to user indicator files.
- **Example:** *UserExternalId1-UserIndicator1, UserExternalId2-UserIndicator2*
- **Mock File Name:** UserIndicator1_Data.csv renamed to UserExternalId1-[timestamp]-[operation].csv and zipped before upload

SFTP_USER_INDICATOR_FILE_MAPPER_DEFAULT_OPERATION

- **Purpose:** Default operation for user indicator files.
 - F = replace file and delete history
 - M = modify file
- **Example:** M Files will be uploaded with the “modify” operation unless specified otherwise.

SFTP_MACRO_INDICATOR_FILE_MAPPER

- **Purpose:** Same mapping logic, applied to macro indicator files.
- **Example:** *MacroExternalId1-MacroIndicator1, MacroExternalId2-MacroIndicator2*
- **Mock File Name:** MacroIndicator1_Stats.csv renamed to MacroExternalId1-[timestamp]-[operation].csv

SFTP_MACRO_INDICATOR_FILE_MAPPER_DEFAULT_OPERATION

- **Purpose:** Default operation for macro indicator files. F means replace the file, and M means modify the file.
- **Example:** F Files will be uploaded with the “replace” operation unless specified otherwise.

✓ Ensure all environment variables are correctly set in *run_ubuntu.bat* before running the application. If you encounter issues, check the log file (log.txt) in the application’s directory for details.

Manual Execution Steps

1. **Edit the Script**
2. **Make the File Executable:**
 - Open the terminal and navigate to the directory containing *run_ubuntu.sh*.
 - Run the following command to ensure it is executable:
`chmod +x run_ubuntu.sh`
3. **Run the Script:**
 - Execute the script:
`./run_ubuntu.sh`
4. **Verify Execution:**



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- The script will check for the presence of the Java runtime and execute SftpClient.jar.

Schedule SftpClient.jar to Run Daily at 05:00 AM

1. Prepare the Script:

- Ensure schedule_ubuntu.sh is in the same directory as run_ubuntu.sh.

2. Run the Scheduler Script:

- Execute schedule_ubuntu.sh to create a cron job:

```
./schedule_ubuntu.sh
```

3. Verify the Task:

- Check the cron jobs to confirm the task is scheduled:

```
crontab -l
```

Remove the Scheduled Task

1. Run the Removal Script:

- Execute remove_schedule_ubuntu.sh to remove the cron job:

```
./remove_schedule_ubuntu.sh
```

2. Verify Removal:

- Check the cron jobs to confirm the task is removed:

```
crontab -l
```

Schedule SftpClient.jar to Run Daily at 05:00 AM

1. Prepare the Script:

- Ensure schedule_ubuntu.sh is in the same directory as run_ubuntu.sh.

2. Run the Scheduler Script:

- Execute schedule_ubuntu.sh to create a cron task:

```
./schedule_ubuntu.sh
```

3. Verify the Task:

- Check if the task is loaded:

```
crontab -l | grep run_ubuntu.sh
```



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Remove the Scheduled Task

1. Run the Removal Script:

- Execute `remove_schedule_ubuntu.sh` to remove the cron task:
`./remove_schedule_ubuntu.sh`

2. Verify Removal:

- Check if the task is unloaded:
`crontab -l | grep run_ubuntu.sh`

Important Notes



- All `.sh` scripts (`run_ubuntu.sh`, `schedule_ubuntu.sh`, `remove_schedule_ubuntu.sh`) must be in the same directory as `SftpClient.jar` and the `jre` folder.
- The folders `user_indicator`, `macro_indicator`, and `portfolio` are created during packaging. You may use custom folders by updating `run_ubuntu.sh`.
- If you encounter errors, check logs in the Task Scheduler or in `log.txt`.

File Structure

Single-User Mode Structure:

/your-directory

```
├─ run_macos.command
├─ schedule_macos.command
├─ remove_schedule_macos.command
├─ SftpClient.jar
├─ README_MACOS.md
├─ README_MACOS.pdf
├─ jre/
├─ user_indicator/    # Contains files to upload
├─ macro_indicator/  # Contains files to upload
├─ portfolio/        # Contains files to upload
```

Multi-User Mode Structure:

/your-directory

```
├─ run_macos.command
├─ schedule_macos.command
├─ remove_schedule_macos.command
├─ SftpClient.jar
├─ README_MACOS.md
```



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```
└─ README_MACOS.pdf
└─ jre/
└─ user_indicator/
|   └─ user1/      # User-specific subdirectory for "user1" SFTP account
|   └─ user2/      # User-specific subdirectory for "user2" SFTP account
|   └─ user3/      # User-specific subdirectory for "user3" SFTP account
└─ macro_indicator/
|   └─ user1/      # User-specific subdirectory for "user1" SFTP account
|   └─ user2/      # User-specific subdirectory for "user2" SFTP account
|   └─ user3/      # User-specific subdirectory for "user3" SFTP account
└─ portfolio/
|   └─ user1/      # User-specific subdirectory for "user1" SFTP account
|   └─ user2/      # User-specific subdirectory for "user2" SFTP account
|   └─ user3/      # User-specific subdirectory for "user3" SFTP account
```

Note: When using multi-user mode, empty folders will be created automatically if they don't exist. Macro and user indicator files placed in the root directory will be automatically distributed to user-specific folders.

Troubleshooting

- **Java Runtime Not Found** Ensure the jre folder is present.
- **Task Scheduler Errors** Confirm you have the required permissions. Check the History tab for details.
- **Incorrect Environment Variables** Double-check the variables in run_ubuntu.sh for typos or invalid values.